Errata closure report

All seven groups from the fresh v1.3.0 errata report have been addressed and pass the documented closure checks. This delivery is the complete corrected 2,160-question bank, with solutions, diagrams, import files and a paper generator.

Group	Correction	Named IDs
E01	Curie-law pair excluded from the same paper	2
E02	Six Difficult tasks replaced with coupled problems	6
E03	2129 now requires the diagram orientation	1
E04	Five malformed explanation clauses removed	5
E05	Three diffraction ordinals corrected	3
E06	Coil article problem corrected throughout its family	5
E07	0192 stem and options form complete sentences	1

Twenty-two named records are changed. The adjacent symbolic coil case 1540 is also corrected, giving 23 changed records. The register has 24 rows: 23 named finding-ID entries, including the 1427 overlap, plus one additional correction.

Parameters retained

Parameter	Final bank
Size and coverage	2,160 four-option MCQs; 20 units; stable IDs
Difficulty	648 Easy / 972 Medium / 540 Difficult
Class and answer letters	1,080 XI / 1,080 XII; 540 each A/B/C/D
Figures	216 linked records / 101 required / 68 unchanged PNGs
Numerical alternates	270; seven synchronized with their current parents
Selection families	1,110 after merging the two Curie-law families

134,234 release assertions passed with no recorded failures. The six replacement tasks have separate printed-input calculations; 1,005 generated papers pass the tested rules. A final ZIP check follows this report and is supplied as the delivery verification.

Six replacement tasks checked separately

E02 is resolved through substantive rewrites at the same IDs, preserving Difficult totals and answer-letter positions. The rationales describe the actual coupled reasoning. Difficulty remains an editorial judgement rather than a student-performance statistic.

ID / key	Verified result	Required reasoning
1015 / C	17.3469 cm	Resolve the first beat branch using two observations, propagate the density change at fixed tension, resolve a second beat branch from its trend, and invert the second-harmonic length relation.
1427 / A	61.1111 V	Link two measured curvatures by retarding work, infer or eliminate the charge-to-mass ratio, and impose the strip tangency condition to obtain the additional energy loss. A ratio solution still requires all three magnetic geometries and the calibration potential.
1548 / A	0.329508 s	Infer both coil parameters, determine a heat-triggered switching current using dissipation sharing, and solve the reversed-source transient with current continuity and a signed target.
1745 / A	4	Combine two integer-order interference conditions, include the plate phase shift, determine the overlap of wavelength-dependent diffraction envelopes, and count only strictly interior positions.
2046 / D	5 mC	Infer resistance and clamp voltage from distinct diode states, establish the switching threshold, and integrate only the conducting portion of both source ramps.
2159 / C	0.653197 mm	Use three bridge configurations to isolate the stretched-wire resistance, connect resistance change to constant-volume deformation, and apply the gauge zero correction before converting the length ratio into diameter.

Independent verification and shortest-path review

The new checker imports no authoring answers or replacement expressions. It enumerates beat branches and optical orders, integrates diode charge and resistor heat, solves current or energy conditions, and eliminates bridge and particle unknowns through separately written equations.

For 1427 the shorter ratio solution was also checked: $\Delta V/600=(0.006^2-0.005^2)/(0.012^2-0.006^2)$. It still uses both calibration curvatures, the retarding potential and the strip geometry. For 2159 the direct fourth-power diameter relation was checked against the equivalent constant-volume length calculation. The published rationales accommodate these valid shorter routes.

Each listed answer set has exactly one option agreeing with the separate computation. The existing numerical alternate of 1015 now asks for the bridge movement in cm: 17.3469388 cm, rounded to 17. Its stored scale, value, integer and parent ID are synchronized.

Selection guard and meaningful figure use

E01 · Curie-law pair

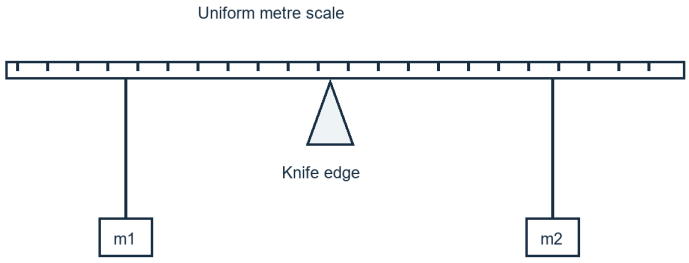
1318 and 1321 now share GLOBAL-CURIE-LAW. The earlier constructed paper contains both and now has only 24 unique families, so it fails the existing unique-family rule. The pair is also included in explicit forbidden-ID checks. All earlier guards remain in force; none of the 1,005 test papers selects this pair together.

E03 · Corrected question 2129

The displayed scale is initially balanced with its knife edge exactly under its own centre of gravity and m1=2m2. The point of suspension of m1 is shifted 1 cm toward the knife edge, remaining on its original side of the pivot. Which shift of m2 restores balance if the knife edge stays fixed? Refer to experimental arrangement 04.

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Experimental arrangement 04



Schematic • not to scale • use the quantities stated in the question

The diagram places m1 to the left. Moving it toward the pivot gives a rightward shift of 1 cm. At fixed pivot, m1 Δx1+m2 Δx2=0 and m1=2m2, so m2 moves 2 cm left: option C. If the two depicted positions are reflected, the same printed inward movement requires m2 to move 2 cm right: option A.

The text alone therefore leaves A and C unresolved; the image supplies the necessary orientation. The existing diagram, its accessible description and figure-required flag remain appropriate. This replaces the old absolute “to the right” wording that made the image dispensable.

Language and export corrections

Group	IDs	Current correction
E04	0255, 0708, 1427, 1840, 2153	Four complete inverse-solution clauses; 1427 has a new full solution. The reported malformed determiner constructions are absent.
E05	1741, 1742, 1743	“Every third” replaces “Every 3th” in all three MCQs and their numerical alternates.
E06	1537, 1538, 1539, 1541, 1542	“a coil with 100 turns” or “a coil with 50 turns” replaces the incorrect indefinite article. Three numerical alternates updated.
A01	1540	Additional symbolic sibling now says “a coil with ξ turns”.
E07	0192	A grammatical question stem and four complete option sentences; B remains the equal-tension answer.

Template and active-record synchronization

The common coil template now uses “a coil with {N} turns”, so its numeric and symbolic substitutions do not require choosing an indefinite article. The diffraction template expresses orders as integer multiples. Current retained candidate snapshots also contain the revised question and explanation text.

Seven numerical alternates updated

1015 has a rebuilt numerical question, value and rounding result. The alternates of 1537, 1538, 1539, 1741, 1742 and 1743 receive the same language correction as their MCQ parent, while retaining their computed values. The other 263 numerical alternates are unchanged.

Complete exports checked

The canonical JSON, JSONL, CSV, questions-only export, answer/explanation export, question index, coverage map and complete student/teacher HTML books were regenerated. Checks compare the printed stems, choices, assumptions, keys and explanations to the canonical bank and confirm figure linkage. No old version of a changed question remains in a current delivery export.

Historical before/after evidence and earlier authoring stages are labelled as such and retained for traceability. They are not alternate current banks.

Exact correction register

All shortened numbers below use JEE27-PHY-. E01-E07 are the seven reported groups. A01 is the additional symbolic coil correction. Every row is resolved in v1.4.0.

Group	ID	Final key	Final answer
E01	1318	A	2.4×10^{-4}
E01	1321	B	inversely proportional to its absolute temperature
E02	1015	C	17.3469 cm
E02	1427	A	61.1111 V
E02	1548	A	0.329508 s
E02	1745	A	4
E02	2046	D	5 mC
E02	2159	C	0.653197 mm
E03	2129	C	Move m2 2 cm to the left
E04	0255	B	0.2
E04	0708	D	10
E04	1427	A	61.1111 V
E04	1840	B	0.5
E04	2153	B	1500
E05	1741	A	0.0001667 m
E05	1742	A	12 mm
E05	1743	B	5
E06	1537	D	0.11 A
E06	1538	C	0.14 C
E06	1539	B	0.1087 J
E06	1541	A	5
E06	1542	A	3
E07	0192	B	The tension has the same magnitude at every point along the string.
A01	1540	D	50

The CSV register additionally lists changed fields and correction details. Complete before/after records are in errata/revision_changes.json and errata/changed_question_evidence.json.

Verification and using the full bank

Verification layer	Result
Current named errata	23 finding-ID closure checks PASS; all seven groups resolved; additional 1540 language check PASS.
Current new calculations	Six separate printed-input computations PASS. Figure orientation counterexample and previous semantic counterexample checks PASS.
Retained mathematical checks	511 original-model recomputations; 260 retained separate-solver computations; 14 v1.2 and 24 v1.3 replacement checks; 54 experimental computations. All PASS.
Numerical delivery	All 270 rounding records checked; parent associations and current export wording checked.
Generated sections	1,005 seeds: 50000-50999 plus 10377, 3391, 830, 27 and 40000. Section sizes, class/difficulty quotas, key balance, unit cap, family uniqueness, prior guards and source/answer agreement PASS.
Prior correction continuity	Earlier v1.3 correction fields remain intact unless explicitly superseded by a verified current correction. Earlier named physical-condition and selection fixes are also rechecked.

Open the complete bank

Extract JEE_Main_Physics_v1.4.0.zip. Open Review_and_Solutions.html for the teacher book or Questions.html for the student book, with the figures folder kept beside them. Both books contain all 2,160 records and work offline. Sample_Paper.html and Sample_Solutions.html contain a 20-MCQ plus 5-numerical Physics section.

Evidence and integrity

The package includes JSON/JSONL/CSV import files, separate student/answer data, a schema, index, coverage map, generator, calculation evidence, this report and the closure register. checksums.sha256 covers the delivered members. The final ZIP was checked separately after assembly; see v1.4.0_delivery_verification.json supplied beside it.

Scope of assurance

The official scope basis remains the NTA JEE Main 2026 syllabus and examination format identified in the supplied source references. The checks close the reported errata and verify preserved parameters. They do not claim a new independent derivation of every bank question, external human subject-expert review or empirical difficulty calibration. Earlier releases are preserved.

Canonical bank SHA-256: d0a5ac908fddd1fb66112727a2cfa3a422e901a5bbab3d7d8fd2727baf5c867d